Unit 4 Guided Notes — Geometry: Angles & Triangles

Name _____ **Date** _____ **Period** _____

Name the relationship, write the equation, solve for x. Angles always add up to something.

🎯 By the end of this unit, I can…

- Use angle relationships formed by parallel lines and a transversal
- Apply the Triangle Angle-Sum and Exterior-Angle Theorems
- Solve for unknown angles by writing and solving equations

Standard: NC.8.G.5

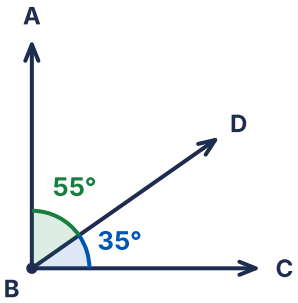
Fill in the blanks as we go. For every example and problem, **write each step on its own line** in the space given, and box your final answer.

1 Angle Relationships

Relationship	Definition	Sum / Rule
Complementary	Two angles that add to _____ °	_____
Supplementary	Two angles that add to _____ °	_____
Vertical Angles	_____	_____
Linear Pair	_____	_____

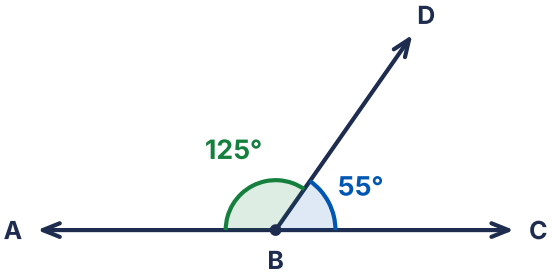
Each angle is named with three letters — the vertex is the MIDDLE letter (e.g. ∠ABD has vertex B). Arcs show which angle each measure belongs to.

COMPLEMENTARY = 90°



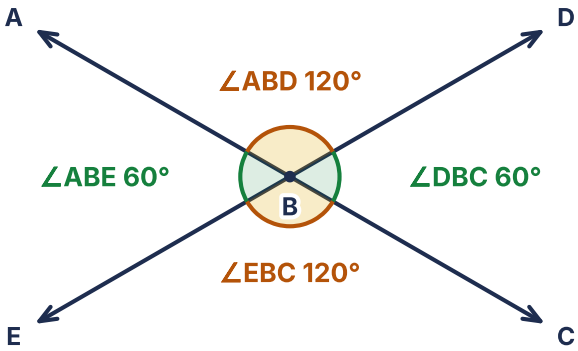
35° + 55° = 90°

SUPPLEMENTARY = 180°



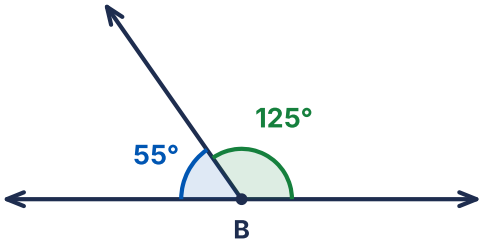
55° + 125° = 180°

VERTICAL ANGLES



Opposite angles are EQUAL: ∠ABD = ∠EBC, ∠DBC = ∠ABE

LINEAR PAIR = 180°



side by side on a line: 125° + 55° = 180°

Quick Check

MC Two angles form a linear pair. One measures 115°. What is the other?

- ○ 65°
- ○ 75°
- ○ 245°

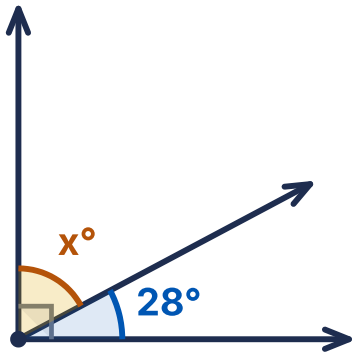
Match the Relationship to its Rule

MATCH **Write the letter of the matching rule on each line.**

_____ 1. Complementary	**A.** Always EQUAL
_____ 2. Supplementary	**B.** Add to 90°
_____ 3. Vertical Angles	**C.** Adjacent angles on a line → 180°
_____ 4. Linear Pair	**D.** Add to 180°

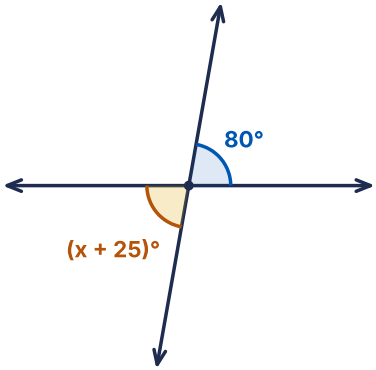
✏ Practice: Angle Relationships

1. ★ Try it The two angles are complementary. Find x.



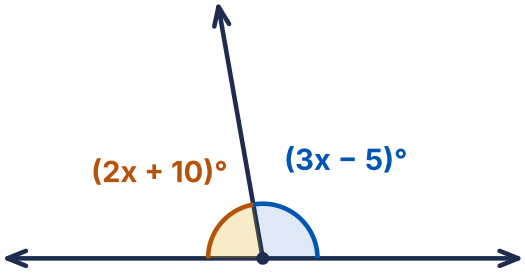
Answer: _____

2. ★★ On level Two lines cross. Find x.



Answer: _____

3. ★★★ Challenge The two angles form a linear pair. Find x.

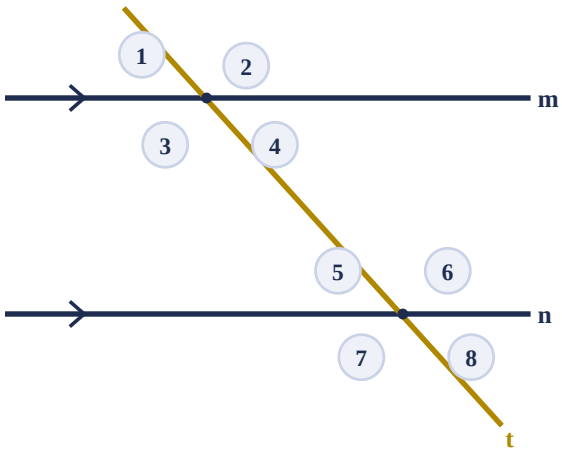


Answer: _____

2 Parallel Lines & Transversals

📌 **REMEMBER:** Corresponding, Alternate Interior, Alternate Exterior, and Vertical angles are CONGRUENT (equal). Same-Side Interior and Same-Side Exterior angles are SUPPLEMENTARY (add to 180°).

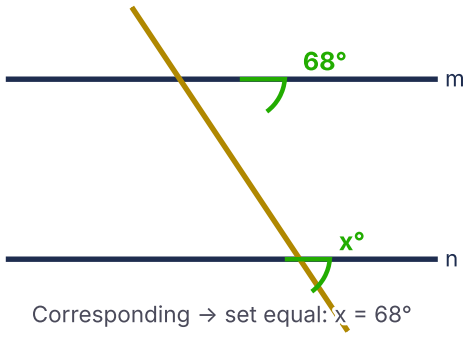
PARALLEL LINES & TRANSVERSAL



Two parallel lines (m ∥ n) cut by transversal t form 8 angles. Corresponding: (1,5)(2,6)(3,7)(4,8) · Alt. Interior: (3,6)(4,5) · Alt. Exterior: (1,8)(2,7) · Same-Side Interior: (3,5)(4,6)

Angle Pair Type	Relationship	Rule to Use	Example
Corresponding	_____	_____	68° → x = 68°
Alternate Interior	_____	_____	101° → x = 101°
Alternate Exterior	_____	_____	142° → x = 142°
Vertical	_____	_____	120° → x = 120°
Same-Side Interior	_____	_____	134° → x = 46°
Same-Side Exterior	_____	_____	77° → x = 103°

CORRESPONDING ANGLES ARE EQUAL



Corresponding → set equal: x = 68°

Worked Example — ∠1 = 3x + 15 and ∠2 = 5x − 7 (alternate interior angles)

`Alternate interior angles are EQUAL:`

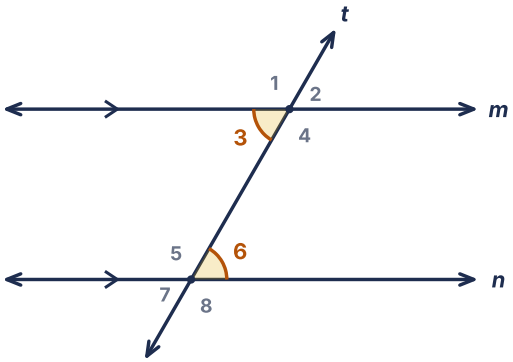
`Subtract 3x:`

`Add 7:`

✓ `Divide by 2:`

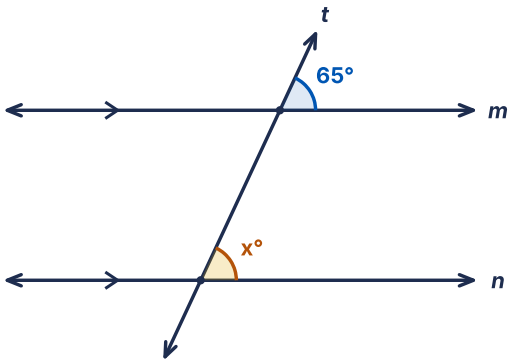
✏ Practice: Parallel Lines & Transversals

1. ★ Try it Lines m and n are parallel. What kind of angle pair are ∠3 and ∠6?



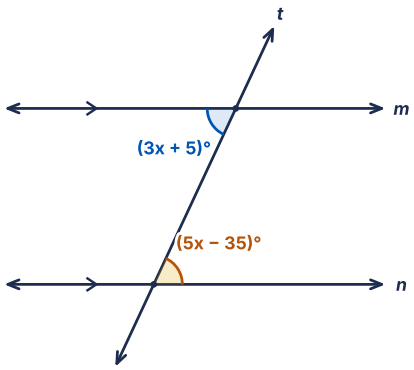
- ○ Corresponding
- ○ Alternate interior
- ○ Same-side interior

2. ★ Try it Lines m and n are parallel. Find x.



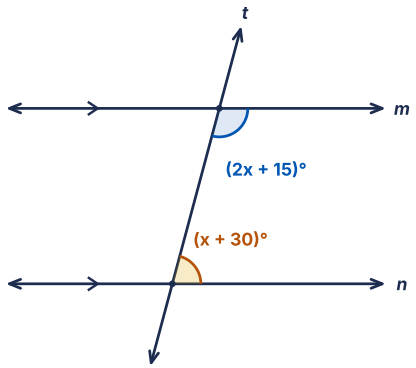
Answer: _____

3. ★★ On level Lines m and n are parallel. Find x.



Answer: _____

4. ★★★ Challenge Lines m and n are parallel. Find x. Then find the measure of the angle marked (2x + 15)°.

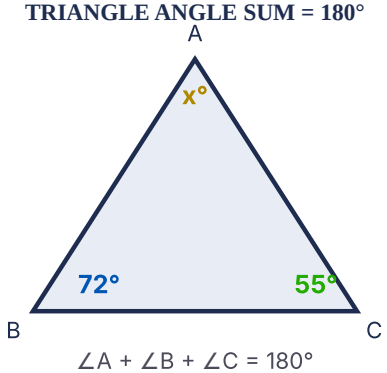


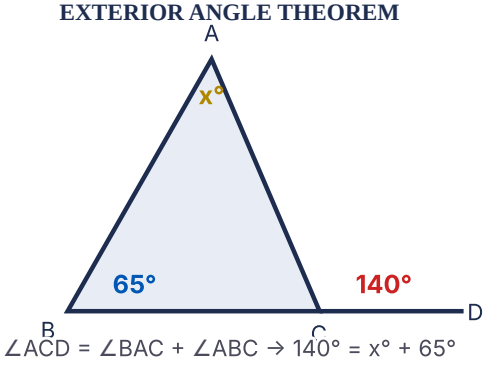
Answer: _____

3 Triangle Theorems

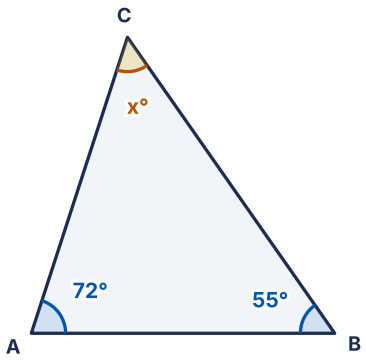
🔑 **TRIANGLE ANGLE SUM:** The three interior angles of a triangle always add up to 180°.

🔑 **EXTERIOR ANGLE THEOREM:** An exterior angle equals the SUM of the two non-adjacent (remote) interior angles.



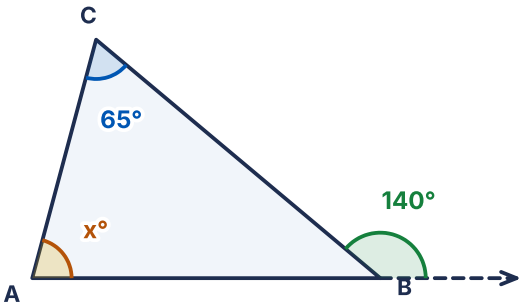


Problem 1 — Triangle Angle Sum: x + 72 + 55 = 180



Solve:

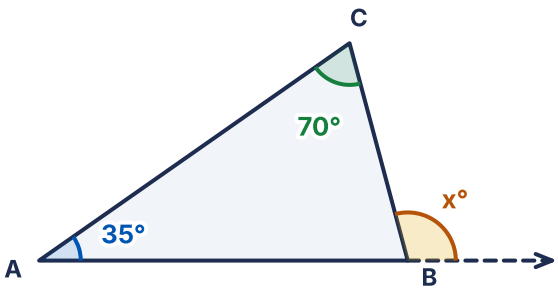
Problem 2 — Exterior Angle Theorem: x + 65 = 140



the outside angle = the two far inside angles

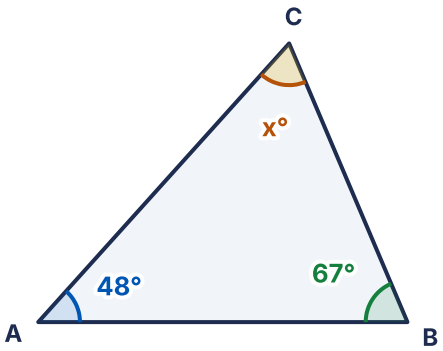
Solve:

ERROR ANALYSIS Jordan's Exterior Angle Mistake — A triangle has interior angles 35° and 70°. Exterior angle = x. Jordan wrote `x = 70 − 35 = 35°`.



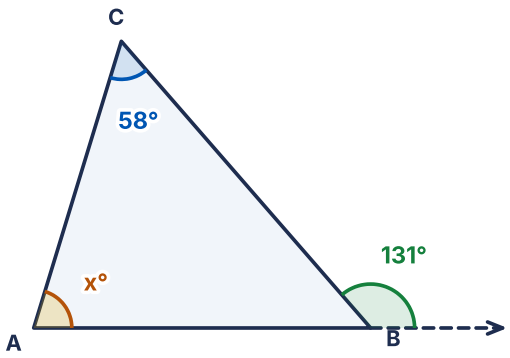
✏ Practice: Triangle Theorems

1. ★ Try it Find x.



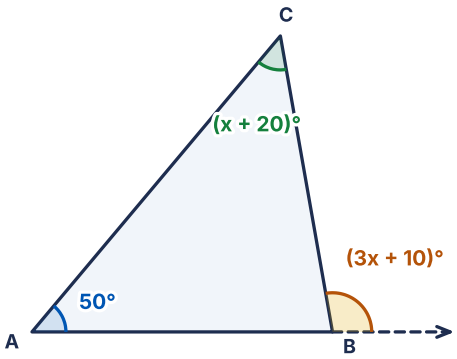
Answer: _____

2. ★★ On level The exterior (outside) angle is 131°. Find x.



Answer: _____

3. ★★★ Challenge Find x. Then find the measure of the exterior angle, (3x + 10)°.



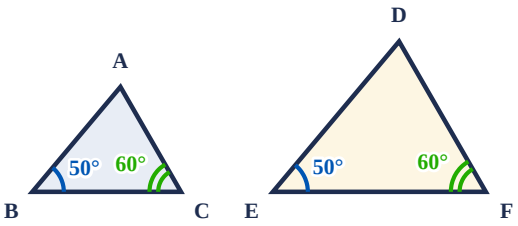
Answer: _____

4 Angle-Angle (AA) Similarity

🔑 **SIMILAR TRIANGLES:** Two triangles are *similar* (written △ABC ~ △DEF) when they have the same shape but not necessarily the same size. Corresponding angles are equal and corresponding sides are in proportion.

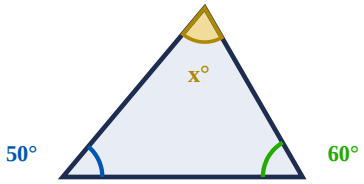
🔑 **ANGLE-ANGLE (AA) CRITERION:** If two angles of one triangle are equal to two angles of another triangle, then the triangles are similar. You only ever need to check TWO pairs of angles.

ANGLE-ANGLE (AA) CRITERION



matching marks = equal angles → △ABC ~ △DEF

WHY TWO ANGLES ARE ENOUGH

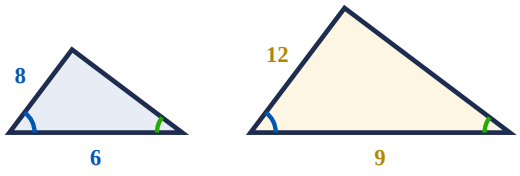


50° + 60° + x° = 180° → x = 70°

the third angle is forced, so all three match

🔑 **WHY TWO ANGLES ARE ENOUGH:** Every triangle's angles add to 180°. Once two pairs of angles match, the third pair has no choice but to match as well — so all three pairs are equal.

SIMILAR → SIDES IN PROPORTION

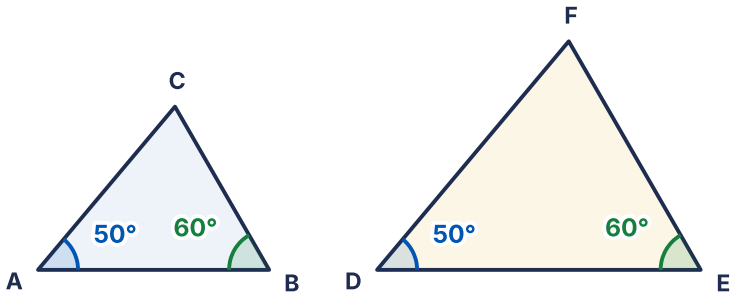


9 ÷ 6 = 1.5 and 12 ÷ 8 = 1.5

scale factor 1.5

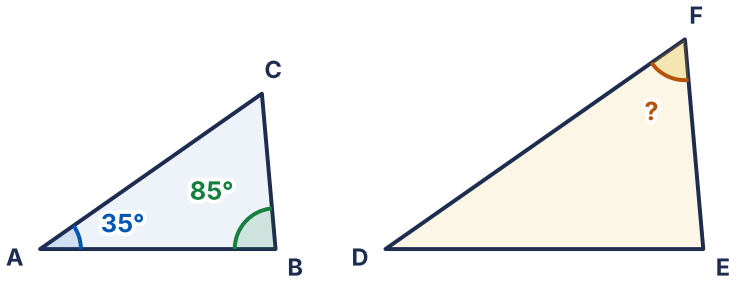
📌 **WATCH OUT:** AA needs two pairs of angles that *match each other* — not simply two angles given in each triangle.

Problem 1 — Are they similar? ∠A = 50°, ∠B = 60° and ∠D = 50°, ∠E = 60°



Decide:

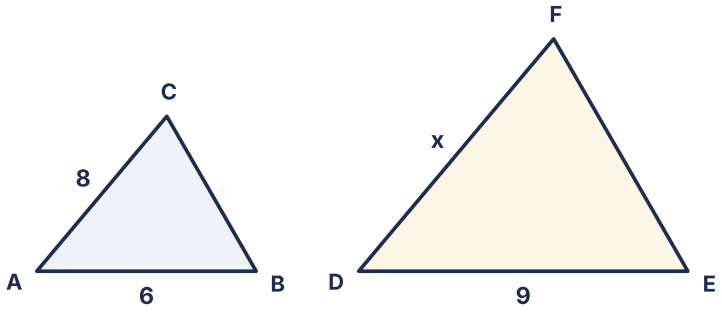
Problem 2 — △ABC ~ △DEF with ∠A = 35° and ∠B = 85°. Find ∠F.



F matches C

Solve:

Problem 3 — Similar triangles: sides 6 and 8 match sides 9 and x. Find x.

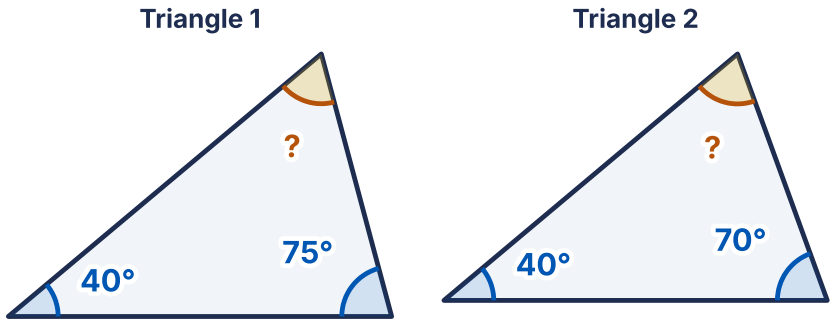


6 → 9 and 8 → x

Solve:

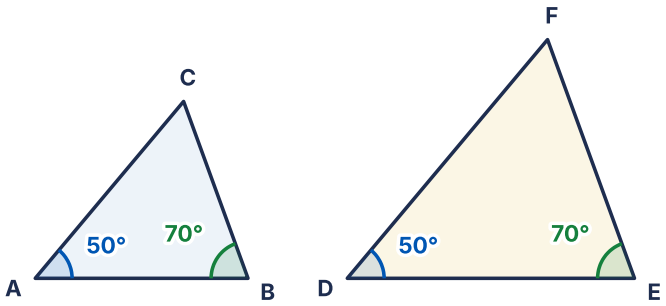
ERROR ANALYSIS Riley's AA Mistake — Triangle 1 has angles 40° and 75°. Triangle 2 has angles 40° and 70°. Riley wrote

> "Each triangle shows two angles, so they are similar by AA."



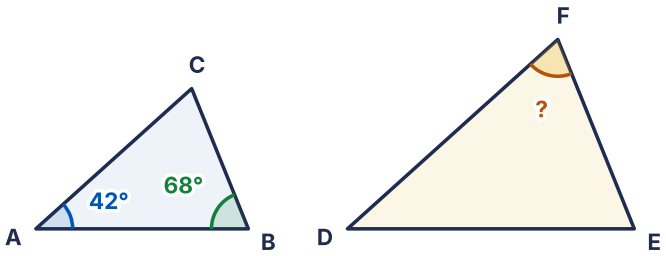
✏ Practice: AA Similarity

1. ★ Try it Are these two triangles similar?



- ○ Yes — two pairs of angles match
- ○ No — they are different sizes
- ○ You can't tell

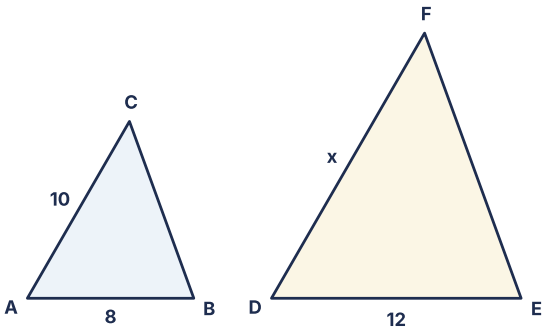
2. ★★ On level △ABC ~ △DEF. Find the measure of ∠F.



F matches C

Answer: _____

3. ★★ On level The triangles are similar. Side 8 matches side 12, and side 10 matches side x. Find x.

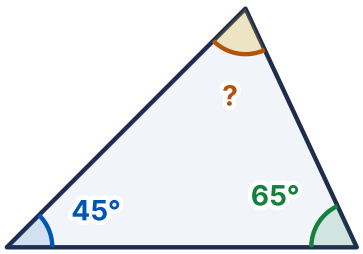


8 → 12 and 10 → x

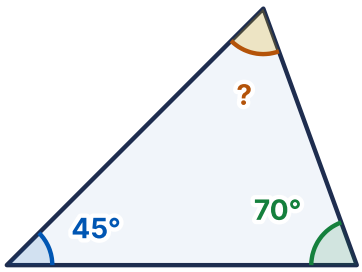
Answer: _____

4. ★★★ Challenge Triangle 1 shows 45° and 65°. Triangle 2 shows 45° and 70°. Sam says they are NOT similar. Is Sam right?

Triangle 1



Triangle 2



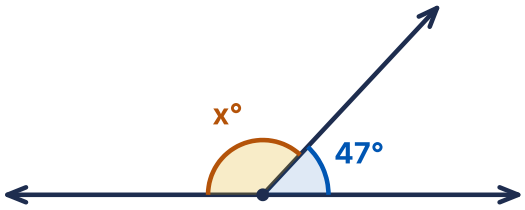
- ○ Yes — they are not similar
- ○ No — they are similar

✏ Part 2 — Comprehensive Practice

📌 **Show every step on its own line.** Box your final answer.

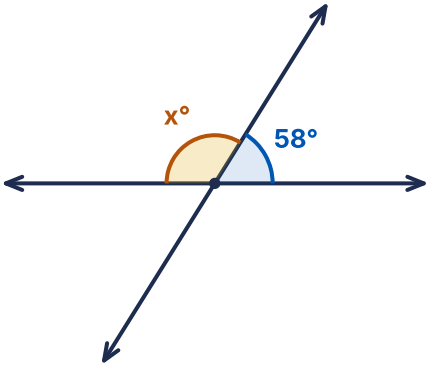
✏ Comprehensive Practice All sections

1. ★ Try it The two angles are supplementary. Find x.



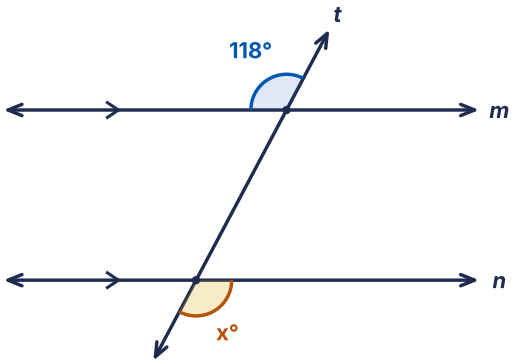
Answer: _____

2. ★★ On level Two lines cross. Find x.



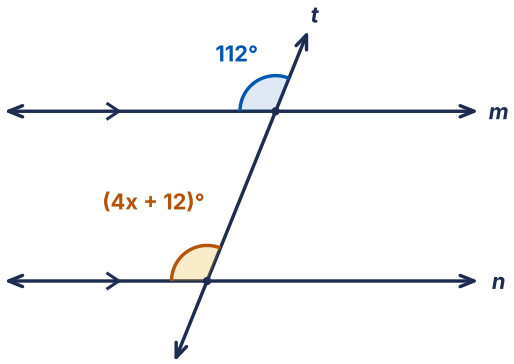
Answer: _____

3. ★ Try it Lines m and n are parallel. Find x.



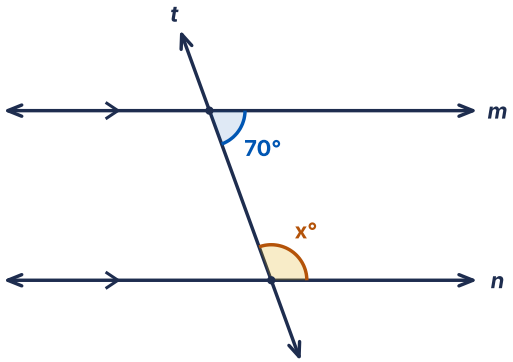
Answer: _____

4. ★★ On level Lines m and n are parallel. Find x.



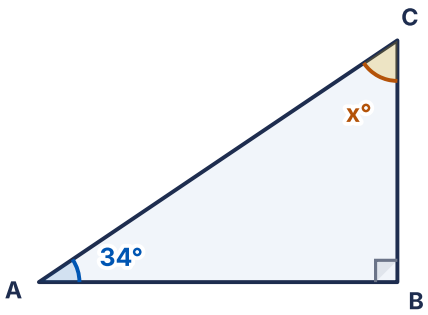
Answer: _____

5. ★★★ Challenge Lines m and n are parallel. Maya says x = 70 because the angles "look the same." Find the correct value of x.



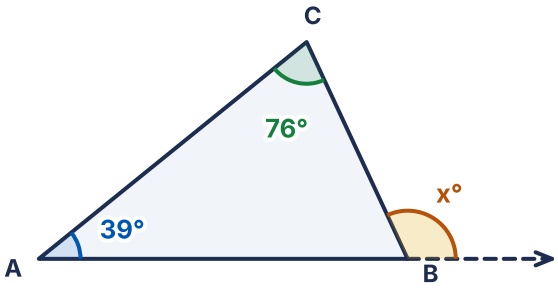
Answer: _____

6. ★ Try it Find x. (The square corner is a right angle.)



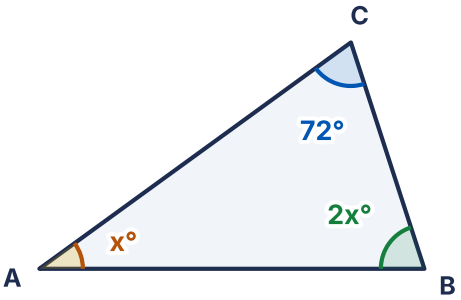
Answer: _____

7. ★ Try it Find the exterior angle x.



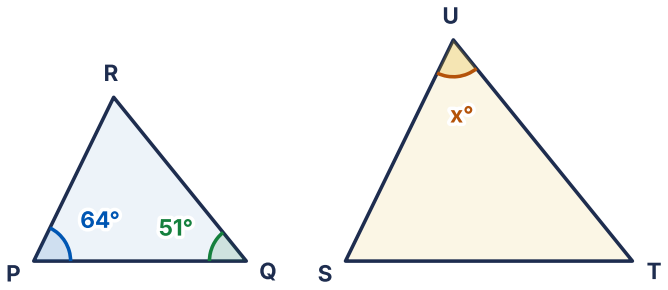
Answer: _____

8. ★★★ Challenge Find x.



Answer: _____

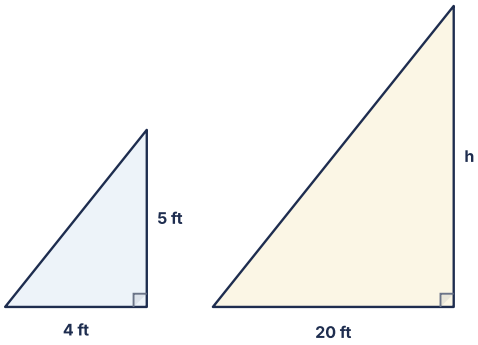
9. ★★ On level △PQR ~ △STU. Find x.



U matches R

Answer: _____

10. ★★★ Challenge A 5 ft student casts a 4 ft shadow. At the same time, a tree casts a 20 ft shadow. The two triangles are similar (AA). How tall is the tree?



student and shadow → tree and shadow (not to scale)

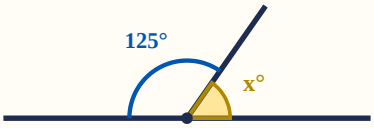
Answer: _____

More Practice by Topic

Extra skill practice for each topic. ★ = Try it, ★★ = On level.

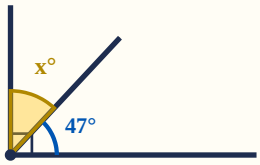
A. Angle Relationships — Solve for x

★ Try it **1) Supplementary Angles, given 125°. x = ?**



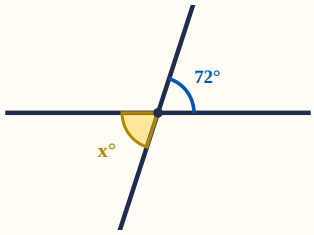
x = _____

★ Try it **2) Complementary Angles, given 47°. x = ?**



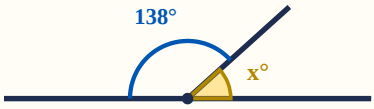
x = _____

★ Try it **3) Vertical Angles, given 72°. x = ?**



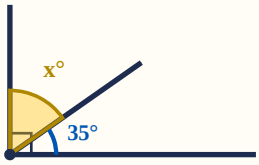
x = _____

★ Try it **4) Linear Pair, given 138°. x = ?**



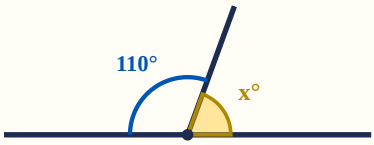
x = _____

★ Try it **5) Complementary Angles, given 35°. x = ?**



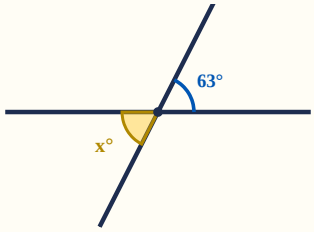
x = _____

★ Try it **6) Supplementary Angles, given 110°. x = ?**



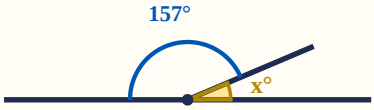
x = _____

★ Try it **7) Vertical Angles, given 63°. x = ?**



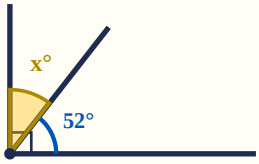
x = _____

★ Try it **8) Linear Pair, given 157°. x = ?**



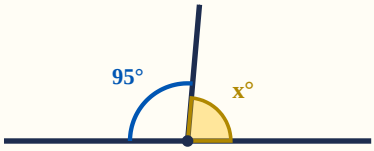
x = _____

★ Try it **9) Complementary Angles, given 52°. x = ?**



x = _____

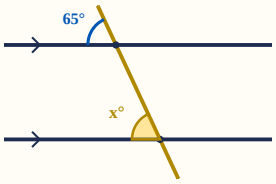
★ Try it **10) Supplementary Angles, given 95°. x = ?**



x = _____

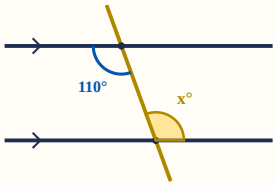
B. Parallel Lines & Transversals

★ Try it **1) Corresponding, given 65°. x = ?**



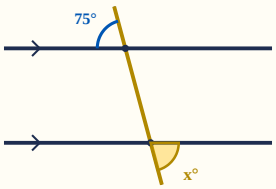
x = _____

★ Try it **2) Alternate Interior, given 110°. x = ?**



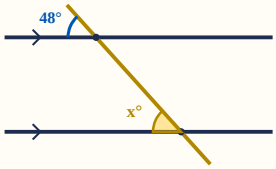
x = _____

★ Try it **3) Alternate Exterior, given 75°. x = ?**



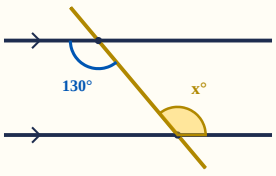
x = _____

★ Try it **4) Corresponding, given 48°. x = ?**



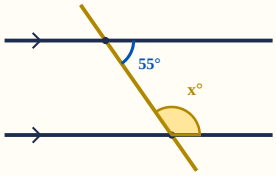
x = _____

★ Try it **5) Alternate Interior, given 130°. x = ?**



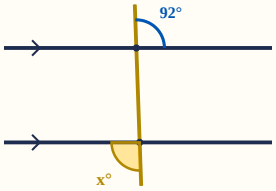
x = _____

★ Try it **6) Co-Interior, given 55°. x = ?**



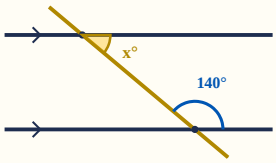
x = _____

★ Try it **7) Alternate Exterior, given 92°. x = ?**



x = _____

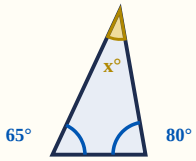
★ Try it **8) Co-Interior, given 140°. x = ?**



x = _____

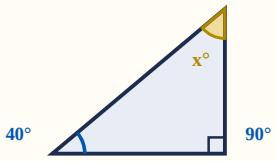
C. Triangle Theorems

★ Try it **1) Angle Sum, given 65°, 80°. x = ?**



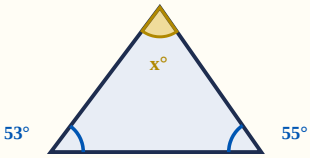
x = _____

★ Try it **2) Angle Sum, given 40°, 90°. x = ?**



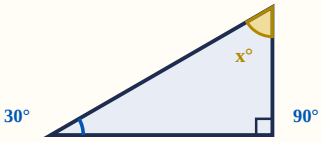
x = _____

★ Try it **3) Angle Sum, given 53°, 55°. x = ?**



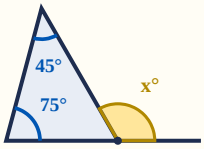
x = _____

★ Try it **4) Angle Sum, given 30°, 90°. x = ?**



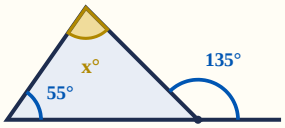
x = _____

★ Try it **5) Exterior Angle Theorem, remotes 45°, 75°. Exterior x = ?**



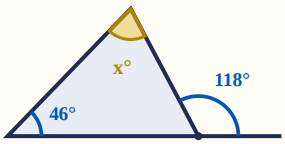
x = _____

★ Try it **6) Exterior Angle Theorem, remote 55°, exterior = 135°. x = ?**



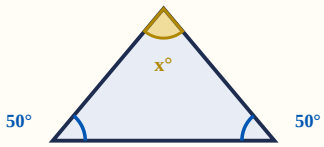
x = _____

★ Try it **7) Exterior Angle Theorem, remote 46°, exterior = 118°. x = ?**



x = _____

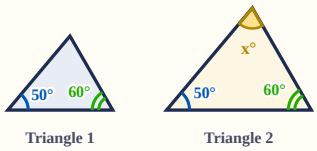
★ Try it **8) Angle Sum, given 50°, 50°. x = ?**



x = _____

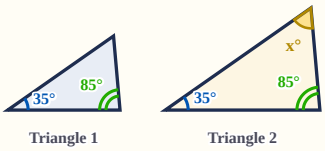
D. Angle-Angle (AA) Similarity

★★ On level 1) AA Similarity — two angles match: 50° and 60°. x = ?



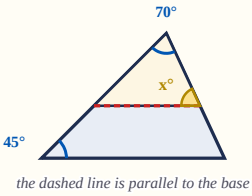
x = _____

★★ On level 2) △ABC ~ △DEF, given 35° and 85°. x = ?



x = _____

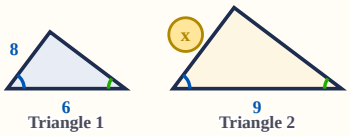
★★ On level 3) Parallel line inside a triangle, given 70° and 45°. x = ?



the dashed line is parallel to the base

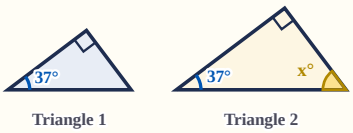
x = _____

★★ On level 4) Similar triangles — sides 6 and 8 match 9 and x. x = ?



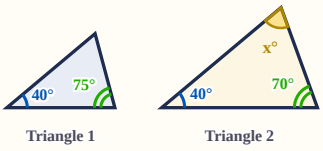
x = _____

★★ On level 5) Similar right triangles, given 37°. x = ?



x = _____

★★ On level 6) Triangle 1: 40°, 75°. Triangle 2: 40°, 70°. Find x, Triangle 2's third angle.



x = _____